



Importance of habitat type classifications for predicting ruffed grouse use of areas for drumming

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ABSTRACT

To incorporate ruffed grouse (*Bonasa umbellus*) habitat planning in forest management, it is necessary for managers to understand factors contributing to grouse habitat use. Previous studies examining ruffed grouse drumming habitat documented relationships between drumming grouse and broad vegetation categories (e.g., northern hardwoods, young aspen [*Populus* spp.], oak [*Quercus* spp.]), but few studies have documented how drumming grouse respond to ecological variations in site conditions of aspen or other vegetation types that might be used. Our objectives were to determine the utility of habitat type classifications in predicting the occurrence of ruffed grouse drumming habitat in the western Upper Peninsula of Michigan, and demonstrate how classifications may be used to understand how forest management may affect ruffed grouse habitat. We used survey routes on state land and conducted drumming surveys during mid-April and early May at 78 points in 2005 and 2006. We recorded the number of drumming males heard at each point, the azimuth to where the grouse was heard, and a qualitative measure of distance to determine in which forest stands grouse were drumming. Using GIS, we determined the specific vegetation type, age class, and habitat type, evaluated habitat suitability, and determined a suitability score for areas in which grouse were drumming. We constructed a logistic regression model that calculated the probability of grouse use of areas for drumming based on vegetation characteristics at used and random locations. Our results indicated that the probability of grouse use of an area for drumming is based on inherent site characteristics (i.e., habitat type) and habitat suitability. The model is useful for planning forest management activities and understanding how grouse may respond to spatial or temporal changes in vegetation through succession or manipulation.

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1. Introduction

Aspen is a valuable timber resource in Michigan and is important for providing wildlife habitat components. Specifically, ruffed grouse use of the aspen vegetation type for food, nesting, and drumming has been well documented in the literature (Bump et al., 1947; Svoboda and Gullion, 1972). As such, a forest management challenge is when and where to manipulate aspen stands or encourage the development of aspen to sustain timber benefits and habitat for grouse or other wildlife.

To incorporate ruffed grouse habitat considerations into forest management plans, it is necessary for managers to understand factors contributing to grouse production. State agencies (e.g.,

Michigan Department of Natural Resources [MDNR]) often use several tools to assess grouse population trends and potential productivity, such as spring drumming surveys. These surveys, however, do not document vegetation characteristics associated with locations of drumming grouse. This information may be critical for understanding how grouse are distributed throughout landscapes during the breeding season and may provide insights as to what ecological factors are contributing to the observed distribution.

Researchers in the Great Lake States (e.g., Gullion, 1977; Kubisiak et al., 1980; McCaffery et al., 1996) have documented higher densities of drumming grouse in 6–25 year-old aspen stands than in other vegetation types and age classes. Other studies have also reported grouse use of older (≥ 25 years) aspen and non-aspen (e.g., balsam fir [*Abies balsamea*], oak, alder [*Alnus* spp.]) vegetation types for drumming (Palmer, 1963; DeStefano and Rusch, 1984; McCaffery et al., 1996). In Michigan, Hammill and Moran (1986) observed drumming grouse in lowland conifer vegetation types. Most studies have documented relationships

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between drumming grouse and broad vegetation categories or cover types (e.g., northern hardwoods, young aspen, oak; Cade and Sousa, 1985; Fearer and Stauffer, 2004), but few studies have documented how ecological classification systems using biotic and biotic components (i.e., habitat types) may describe use by drumming grouse. Habitat types are geographic areas with similar ecological characteristics that support the same successional trajectory (Daubenmire, 1966). Habitat type classifications group communities and their environments into categories useful for management interpretation (Kotar and Burger, 2000). These classifications allow a better understanding of potential successional trajectories and the distribution of ecological communities that reflect inherent site capabilities, disturbance patterns, and potential response to management.

Aspen is an early successional vegetation type occurring in several different habitat types which are supported by soil types ranging from poorly-drained loams to well-drained sand (Coffman et al., 1980). The structure and composition of aspen varies within habitat type and provides different wildlife habitat components (Felix et al., 2007; Felix, 2008). For instance, aspen in habitat types characterized by dry sandy soils (e.g., Rubicon, Kalkaska series) may contain an oak or pine component whereas aspen stands in habitat types characterized by loamy soils (e.g., onaway series) may contain a significant maple component. Therefore, it is reasonable to expect that being able to classify areas by habitat type, as well as vegetation type (i.e., aspen or the current vegetation community), might be an effective way to predict grouse drumming habitat.

The utility of habitat type classifications for understanding wildlife habitat potential and population demographics has only recently been investigated as a tool to facilitate development of ecologically-based management plans and predict wildlife population response (Haufler et al., 1996; Roloff and Haufler, 1997; Felix et al., 2007). Habitat types may be used to predict wildlife distribution and habitat potential throughout space and time because they have unique and predictable ecological characteristics and successional trajectories (Kotar and Burger, 2000). To design long-term sustainable timber harvest plans that benefit ruffed grouse, it may be beneficial to use habitat type classifications in conjunction with drumming surveys to assess grouse use of aspen and other vegetation types in different habitat types and age classes. Such information may provide insights for understanding landscape-level characteristics affecting grouse drumming distributions, and identifying specific habitat types, and successional stages within them that may be valuable in influencing grouse productivity. Specifically, understanding grouse use of vegetation within different habitat types and seral stages may indicate how drumming habitat availability might change with proportional changes in availability of vegetation in different habitat types or age classes from timber harvesting or succession. Therefore, our objectives were to determine the utility of habitat types in predicting the occurrence of ruffed grouse drumming habitat, and demonstrate how habitat type classifications may be used to facilitate development of ruffed grouse management plans.

2. Study area

Our study area was approximately 800 km² and located in the western Upper Peninsula (UP) of Michigan, which included parts of Dickinson and Marquette counties. Sites selected for assessment were located within the Escanaba River State Forest and the Copper Country State Forest. These areas are managed for timber production and wildlife habitat. This region of Michigan accumulated 93 cm of precipitation annually and averaged 4.2 °C (National Weather Service, <http://www.crh.noaa.gov/mqt/normals/marquette.php>).

Soils in the western UP were dominantly spodosols and bedrock rich in iron.

Well-drained sands and loams typically support hardwood forests dominated by maple (*Acer* spp.) in climax stages. Spruce (*Picea* spp.) and balsam fir occurred frequently as components of hardwoods growing on sandy soils. Other sub-dominant species in hardwoods on loamy soils included basswood (*Tilia americana*), ironwood (*Carpinus caroliniana*), or American elm (*Ulmus americana*). Dry, sandy soils support oak, white pine (*Pinus strobus*), and red pine (*Pinus resinosa*). Poorly-drained areas were interspersed throughout the region and were typically dominated by northern-white cedar (*Thuja occidentalis*), spruce, fir, and tamarack (*Larix laricina*; Albert et al., 1986). Early successional and aspen-dominated forests were also interspersed throughout much of the area (Sommers, 1977).

3. Methods

3.1. Habitat type classification

We spatially delineated boundaries of 10 habitat types as defined by a habitat type classification guide by Coffman et al. (1980) (Fig. 1) using ArcGIS 9.2. Each habitat type was characterized by soil moisture and texture, landform (e.g., outwash plain, moraine), and ecoregion data acquired from the MDNR. Boundaries of habitat types were validated with spatial information (IFMAP; Integrated Forest Monitoring and Assessment Prescription; MDNR, 2003), and 70 random points ground-truthed during a summer field season in 2004. For validation, we ensured that the overstory vegetation and herbaceous layer in our defined habitat type boundaries corresponded with the Coffman et al. (1980) classification.

Habitat types were named according to the typical dominant overstory and understory species in the climax stage that occurred on various soil types (Coffman et al., 1980). Habitat types characterized by xeric, excessively-drained sandy soils included: *Pinus-Vaccinium*-spp. (PV; pine-blueberry), *Quercus-Acer-Epigaea* (QAE; oak-maple-trailing arbutus), and *Acer-Quercus-Vaccinium* (AQV; maple-oak-blueberry). Mesic habitat types characterized by well-drained soils included: *Acer-Tsuga-Maianthemum* (ATM; maple-hemlock-wild lily-of-the-valley), *Acer-Tsuga-Dryopteris* (ATD; maple-hemlock-spinulose shield fern), *Acer-Viola-Osmorhiza* (AVO; maple-violet spp.-sweet cicely), and *Acer-Osmorhiza-Caulophyllum* (AOC; maple-sweet cicely-blue cohosh) habitat types.

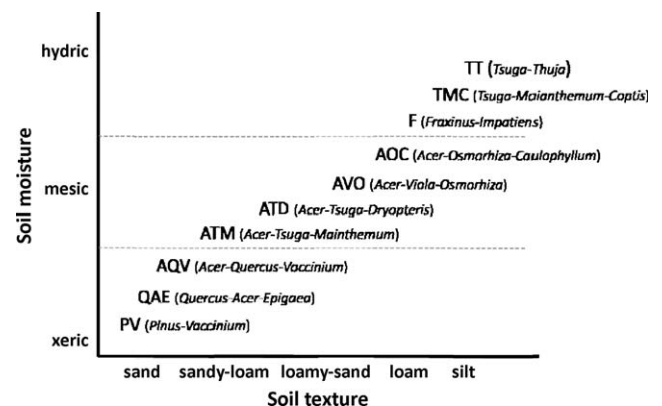


Fig. 1. Relative soil moisture and texture characteristics of habitat types occurring in a 800 km² area in the western Upper Peninsula of Michigan. Habitat types are based on a classification by Coffman et al. (1980). Habitat types are named by dominant overstory and understory species with the strongest tendency to dominate the community in the absence of disturbance. All habitat types are capable of supporting aspen in early successional stages.

Habitat types on poorly-drained soils included: *Tsuga-Maianthemum-Coptis* (TMC; hemlock-wild lily-of-the-valley-goldthread), *Fraxinus*-spp. (F; ash), and *Tsuga-Thuja* (TT; hemlock-cedar).

3.2. Ruffed grouse drumming surveys

We identified locations of forested areas on state land and identified specific GPS coordinates for 78 points on 10–15 routes at least 10 km long for grouse drumming surveys based on stand accessibility in a 4-wheel drive truck and to maximize the number of stands assessed during the drumming survey. All routes collectively contained stands with representation of vegetation in all available vegetation and habitat types within the landscape. A Chi-square test was used to compare representation of vegetation in sampled areas to that of the landscape.

Grouse drumming surveys occurred between May 3–16 in 2005 and April 17–26 in 2006, which corresponded with the peak drumming period in the UP (April 20–May 10). Surveys were conducted according to MDNR protocol for state ruffed grouse drumming surveys (R. Atkinson, MDNR, personal communication). To minimize effects of weather conditions on drumming activity or audibility, surveys were only conducted when the temperature was between -4 and $+4$ °C with no precipitation and when wind was <19 kph. We used the Beaufort wind scale to monitor wind speed, which indicates when small twigs are in constant motion, wind speed is 13–19 kph (NOAA, 2007). Surveys began 30 min before sunrise (approximately 0600 h EDT) and ended around 0900 h EDT (Petraborg et al., 1953).

At each stop, the observer walked at least 15 m away from the vehicle and listened for 4 min, which is the average drumming interval for ruffed grouse (Petraborg et al., 1953). We recorded the number of drumming males heard at each sampling point, the compass bearing from the sampling point to the direction where the drumming was heard, and a qualitative measure of distance: close, medium, or far. Each distance category represented approximately one-third of the radius of audibility; 200 m was the radius of audibility for drumming ruffed grouse in similar vegetation types in Minnesota (Petraborg et al., 1953). “Close” drumming occurred within 60 m of the sampling point, drumming at a “medium” distance occurred 60–130 m from the sampling point, and “far” was 130–200 m. On 15 occasions (i.e., 5 identified for each distance category), the drumming log was located to validate distance assignments and determine directional error. We assumed vegetation type did not influence detection probability of drumming grouse (Zimmerman, 2006). Directional error was the difference between the estimated and actual compass bearing from the survey point to the drumming log.

We used information from the Michigan Operations Inventory Forest Database in ArcGIS 9.2 to determine the proportion of young (0–29 years), medium (30–49 years), and old (≥ 50 years) aspen stands, and the proportion of area in non-aspen vegetation types within each 200 m radius circle. The specified vegetation classes represented those found to be frequently used by drumming grouse (i.e., 6–25 year-old aspen; Gullion, 1977; Kubisiak et al., 1980; McCaffery et al., 1996), as well as other classes in which drumming has not been extensively documented in the Great Lakes region. We then used our spatially-explicit habitat type map to determine the composition of habitat types within each circle. Next, we determined in which aspen age class (young, medium, and old), vegetation type (aspen and non-aspen), and habitat type drumming occurred by using the Distance/Azimuth Tools Extension for ArcView 3.x (Jenness, 2005). This tool established a point at our specified distance (i.e., the midpoint of distance categories) and directional bearing from a point of origin. The specific composition of non-aspen vegetation types was determined using the habitat type classification (Fig. 1) and current landcover maps (MDNR,

2003). We assumed the grouse were drumming in the specific vegetation type, age class, and habitat type occurring at the point established with the tool.

To assess accuracy of the estimated drumming locations, we determined average patch size of age class and habitat type boundaries within each distance category. We used the directional error and accuracy of estimated distance to drumming log locations to determine overall error in estimating vegetation characteristics in which grouse were drumming.

To compare used areas with random sites, at survey points where no drumming occurred, a point was randomly established using the “Sampling Tools” within Hawth’s Analysis Tools for ArcGIS (Beyer, 2004). These random points were placed within the 200 m radius of audibility buffer. Vegetation type, age class, and habitat type were recorded.

3.3. Habitat suitability

We developed a map of ruffed grouse habitat suitability by applying an existing habitat suitability index (HSI) model (i.e., Hammill and Moran, 1986) to the study area. Five habitat variables were quantified: (1) equivalent stem density (i.e., which incorporates vertical structure provided by deciduous, conifer, and shrub stems), (2) stem height, (3) ground to canopy height of conifers, (4) overstory stem height, and (5) interspersed between winter food and fall-to-spring cover (Hammill and Moran, 1986). Structural and compositional characteristics of vegetation were determined for different aspen age classes within different habitat types during the 2005 and 2006 field seasons. Structural and compositional characteristics of non-aspen vegetation types within different habitat types (e.g., northern hardwoods, oak, pine, and lowland conifer) were determined from existing literature (i.e., Rudolf, 1951; Burns and Honkala, 1990; Hanaburgh, 2001).

3.4. Logistic regression

To assess the ability of habitat type classifications in predicting occurrence of ruffed grouse drumming habitat, we used ProcLogistic in SAS software, version 9.2 (SAS Institute Inc., Cary, NC) to create logistic regression models based on the response variable of drumming presence (drumming observed = 1; no drumming = 0), and the explanatory variables of habitat type, age class, and habitat suitability. We tested for differences in grouse use of vegetation classes between years using a Chi-square test. Differences were evident ($\chi^2_{0.05,5} = 90.77$; $p < 0.001$) in the proportional use of vegetation classes, but not in the types of vegetation used. Therefore, we pooled data between years to capture the variability in grouse use of vegetation classes for drumming.

Habitat type (HT) was a class variable, and included 10 types (Table 1). Vegetation class (VEGCLASS) was a class variable and included young aspen (0–29 years), medium-aged (30–49 years), old aspen (>50 years), and non-aspen (Table 1). Habitat suitability index (HSI) was a continuous variable and included values 0–100; 100 = high habitat suitability. The logistic model that we used to estimate probability of drumming habitat use was as follows:

$$\hat{y} = \frac{\exp(\hat{u})}{1 + \exp(\hat{u})},$$

where $\hat{y}$ was the estimated probability an area would be suitable and used for drumming and $\hat{u} = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{\beta}_3 x_3$ was the usual multiple linear regression equation with $x_{1i} = HT_{1-10}$, $x_{2i} = VC_{1-4}$, and $x_3 = HSI$. The suite of models examined included the null model and all possible combinations of HT, VEGCLASS, and HSI. A receiver operating characteristic (ROC) curve was created to assess predictive power. We used Akaike’s Information Criterion to determine which models were valid and to select the best model.

Table 1Percent of different aspen age classes and non-aspen vegetation types within habitat types in a 800 km² area in the western Upper Peninsula of Michigan.

Habitat type	Vegetation class				Dominant overstory composition of non-aspen vegetation class	Total
	Young aspen	Medium aspen	Old aspen	Non-aspen		
PV	0.16	0.09	0.03	3.60	Red pine, jack pine	3.88
QAE	1.20	0.40	0.38	3.62	Oak, red maple	5.60
AQV	1.99	0.61	0.82	6.80	Red maple, oak	10.22
ATM	1.80	0.49	0.52	5.71	White spruce, balsam fir, red maple	8.52
ATD	0.03	0.01	0.07	0.25	Maple spp., hemlock	0.36
AVO	4.24	1.48	1.43	14.10	Maple spp., basswood, elm	21.25
AOC	1.72	0.66	0.76	7.43	Maple spp., basswood, elm	10.57
F	0.59	0.07	0.09	5.06	Ash spp., maple spp.	5.81
TMC	1.02	0.39	0.66	2.60	Hemlock, maple spp., spruce, fir, cedar	4.67
TT	1.02	0.40	0.56	27.14	Spruce, fir, cedar	29.12
Total	13.77	4.60	5.32	76.31		100.00

Young aspen was <30 years old, medium aspen was 30–49 years old, and old aspen was ≥50. Non-aspen vegetation was determined from land cover classification and from the specific habitat type in which it occurred. Habitat types were named according to the dominant overstory and understory vegetation in climax stages. PV = *Pinus-Vaccinium*-spp., QAE = *Quercus-Acer-Epigaea*, AQV = *Acer-Quercus-Vaccinium*, ATM = *Acer-Tsuga-Maianthemum*, ATD = *Acer-Tsuga-Dryopteris*, AVO = *Acer-Viola-Osmorhiza*, AOC = *Acer-Osmorhiza-Caulophyllum*, F = *Fraxinus*-spp., TMC = *Tsuga-Maianthemum-Coptis*, and TT = *Tsuga-Thuja*-spp.

4. Results

The proportion of vegetation classes sampled was slightly different compared to landscape representation ($\chi^2_{0.05,39} = 67.42$; $p = 0.009$), but the proportional differences only deviated an average of $2.4 \pm 0.60\%$. Thus, we assumed our sampled areas were a suitable representation of the entire landscape composition. Methods used to determine the grouse drumming locations were credible. Differences between estimated and actual drumming locations were 24.43 ± 1.01 m for the “close” distance category, 22.57 ± 3.38 m for the “medium” category, and 47.06 ± 4.63 m for the “far” category. Average patch size of vegetation classes was 0.64 ha, 1.08 ha, and 1.32 ha for “close,” “medium,” and “far” distance categories, respectively. Circular areas with a radius equivalent to the upper confidence limit of the error measurements were smaller than the average patch size of vegetation classes and HTs within distance categories. All actual drumming locations occurred in the same habitat type as was determined from the estimated positions. On one occasion (in the “far” distance category), the actual drumming log occurred in a different vegetation class than was determined from the estimate. Thus, predictions of vegetation classes and habitat types of grouse estimated to be drumming <130 m from the observer had a 0% error; those estimated to occur >130 m from the observer had an error of 20% for identifying actual vegetation classes, but not habitat type, in which actual drumming occurred.

The proportion of grouse using aspen was greater than the proportion using non-aspen vegetation types and also greater than the proportion of random points in aspen (Table 2). The number of

drumming grouse in young and old aspen was higher (4.04% and 4.52%, respectively) than the number of random points in each of those vegetation classes; however a higher percentage of random points occurred in medium-aged aspen than did drumming grouse (Table 2). Within HTs, the ATM, AVO, and TT habitat types supported a greater percentage of drumming grouse than the random points (15.48%, 9.76%, 0.72%, respectively). The AQV, QAE, and AOC types had proportionally fewer drumming grouse than random point locations (−20.71%, −2.62%, and −2.62%, respectively; Table 2). The greatest difference between percentages of grouse using vegetation classes compared to random points occurred in young aspen within the AQV habitat type (used = 3.81%, random points = 14.29%), medium aspen within the AVO habitat type (used = 13.33%, random points = 0%), and non-aspen within the AQV habitat type (used = 3.81%, random points = 14.29%; Table 2).

Ruffed grouse drumming habitat suitability was a function of vegetation structure (variables 1–4) and interspersed (variable 5). Vegetation structure was an inherent characteristic of habitat types; different habitat types have different potentials to provide vegetation structure necessary for drumming. Areas with vegetation structure classified as high, medium, and low habitat suitability were not evenly distributed among habitat types or vegetation classes (Fig. 2). The AQV, AVO, AOC, TMC, and TT habitat types supported vegetation structure with drumming habitat suitability > 0 in all aspen age classes and non-aspen vegetation types, whereas the PV, QAE, ATM, ATD, and F habitat types supported suitable drumming habitat in some aspen age classes,

Table 2Percent of grouse using different aspen age classes and non-aspen vegetation types within habitat types in an 800 km² area in the western Upper Peninsula of Michigan compared to percent of random points within each vegetation class.

Habitat type	Vegetation class								Total	
	Young aspen		Medium aspen		Old aspen		Non-aspen		Used	Random
	Used	Random	Used	Random	Used	Random	Used	Random		
QAE	3.81	14.29	2.86	3.57	0.95	0.00	0.95		0.95	3.57
AQV			5.71		8.57	3.57	3.81	14.29	11.43	32.14
ATM	4.76		13.33		6.67	14.29	4.76	3.57	19.05	3.57
AVO	2.86		5.71	21.43	10.48	7.14	4.76	3.57	27.62	17.86
AOC	8.57		1.90	7.14	2.86		4.76		29.52	32.14
TT	1.90	3.57					4.76		11.43	10.71
Total	21.90	17.86	29.52	32.14	29.52	25.00	19.05	25.00	100.00	100.00

Young aspen was <30 years old, medium aspen was 30–49 years old, and old aspen was ≥50. Non-aspen vegetation was determined from land cover classification and from the specific habitat type in which it occurred. Habitat types were named according to the dominant overstory and understory vegetation in climax stages. QAE = *Quercus-Acer-Epigaea*, AQV = *Acer-Quercus-Vaccinium*, ATM = *Acer-Tsuga-Maianthemum*, AVO = *Acer-Viola-Osmorhiza*, AOC = *Acer-Osmorhiza-Caulophyllum*, and TT = *Tsuga-Thuja*-spp.

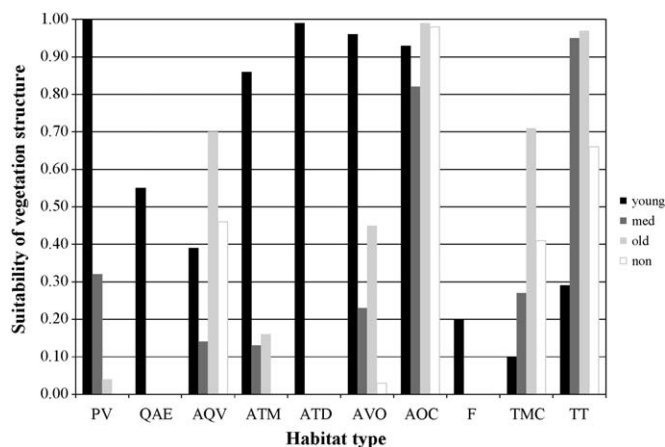


Fig. 2. Habitat suitability index scores within aspen age classes (young, medium, and old) and non-aspen vegetation types occurring within different habitat types in the western Upper Peninsula of Michigan. Young aspen was <30 years old, medium aspen was 30–49 years old, and old aspen was ≥50. Non-aspen vegetation was determined from land cover classification and from the specific habitat type in which it occurred. Habitat types were named according to the dominant overstory and understory vegetation in climax stages. PV = *Pinus-Vaccinium*-spp., QAE = *Quercus-Acer-Epigeaea*, AQV = *Acer-Quercus-Vaccinium*, ATM = *Acer-Tsuga-Maianthemum*, ATD = *Acer-Tsuga-Dryopteris*, AVO = *Acer-Viola-Osmorhiza*, AOC = *Acer-Osmorhiza-Caulophyllum*, F = *Fraxinus*-spp., TMC = *Tsuga-Maianthemum-Coptis*, and TT = *Tsuga-Thuja*-spp.

but not in non-aspen vegetation types (Fig. 2). The mesic-loamy AOC habitat type supported high habitat suitability (>0.80) throughout all aspen age classes and non-aspen vegetation types. Young age classes in 6 of 10 habitat types had the highest suitability among all vegetation classes within habitat types. In some habitat types (i.e., AQV, AOC, TMC, and TT), higher habitat suitability occurred in older aspen age classes than in other vegetation classes. Drumming habitat suitability was never highest in non-aspen vegetation types (Fig. 2).

Areas classified as high habitat suitability (HSI > 75) had a higher percentage of drumming grouse than where random points were located (21.90%), whereas the percentage of grouse drumming in areas classified as low (HSI = 0–39) and medium (HSI = 40–74) habitat were lower (–13.10% and –8.18%, respectively) than the percentage of random points falling in those areas (Table 3); however some vegetation classes with low habitat suitability had a higher percentage of drumming grouse than random points (Table 3). Several vegetation classes classified as medium habitat suitability supported a higher percentage of drumming grouse than random points, but the differences were <2.86% (Table 3).

In the logistic regression analysis, model 1 (Table 4), which contained HT and HSI scores, was the best model. No other models were considered competing, as ΔAIC were all >2 (Burnham and Anderson, 2002). Model parameter estimates were $\beta_0 = 0.19$ (SE = 0.29), $\hat{\beta}_{1,QAE} = -0.89$ (SE = 1.04), $\hat{\beta}_{1,AQV} = -0.91$ (SE = 0.41), $\hat{\beta}_{1,ATM} = 1.77$ (SE = 0.67), $\hat{\beta}_{1,AVO} = -0.20$ (SE = 0.37), $\hat{\beta}_{1,AOC} = 0.15$

Table 3
Percent of drumming grouse using areas classified as low, medium, and high habitat suitability in the western Upper Peninsula of Michigan (2005–2006) compared to percent of random points within each suitability category.

Vegetation class	Low suitability		Medium suitability		High suitability		Total	
	Used	Random	Used	Random	Used	Random	Used	Random
QAE								
Med aspen		3.57					0.00	3.57
Non-aspen	0.95						0.95	0.00
AQV							0.00	0.00
Young aspen	2.86	10.71	0.95	3.57			3.81	14.29
Med aspen	2.86						2.86	0.00
Old aspen	0.95	3.57					0.95	3.57
Non-aspen	0.95	14.29	2.86				3.81	14.29
ATM							0.00	0.00
Young aspen	0.95		1.90		1.90		4.76	0.00
Med aspen	3.81						3.81	0.00
Old aspen	8.57						8.57	0.00
Non-aspen	1.90	3.57					1.90	3.57
AVO							0.00	0.00
Young aspen					2.86		2.86	0.00
Med aspen	11.43	17.86	0.95				12.38	17.86
Old aspen	5.71	3.57		10.71			5.71	14.29
Non-aspen	4.76	3.57					4.76	3.57
AOC							0.00	0.00
Young aspen	3.81		1.90		2.86		8.57	0.00
Med aspen	1.90		1.90	3.57	4.76		8.57	3.57
Old aspen	2.86	3.57	3.81	3.57	3.81		10.48	7.14
Non-aspen	0.95			3.57	3.81		4.76	3.57
TT							0.00	0.00
Young aspen	0.95	3.57	0.95				1.90	3.57
Med aspen					1.90		1.90	0.00
Old aspen	1.90						1.90	0.00
Non-aspen	3.81	7.14	0.95				4.76	7.14
Total	61.90	75.00	16.19	25.00	21.90		100	100

Vegetation class included young aspen (<30 years old), medium aspen (30–49 years old), old aspen (≥50 years old), and non-aspen within different habitat types. Non-aspen vegetation was determined from land cover classification and from the specific habitat type in which it occurred. Habitat types were named according to the dominant overstory and understory vegetation in climax stages. QAE = *Quercus-Acer-Epigeaea*, AQV = *Acer-Quercus-Vaccinium*, ATM = *Acer-Tsuga-Maianthemum*, AVO = *Acer-Viola-Osmorhiza*, AOC = *Acer-Osmorhiza-Caulophyllum*, and TT = *Tsuga-Thuja*-spp.

Table 4

All possible models and their parameters, number of parameters (K), Akaike's Information Criterion (AIC) values, and relative AIC values (ΔAIC) for predicting ruffed grouse use of areas for drumming in the western Upper Peninsula, Michigan (2005–2006).

Model	Parameters	K	AIC	ΔAIC
1	HT, HSI	7	188.82	0.00
2	HT, VEGCLASS, HSI	10	192.67	3.85
3	HSI	2	194.37	5.55
4	VEGCLASS, HSI	5	197.62	8.80
5	HT	6	198.36	9.55
6	HT, VEGCLASS	9	202.25	13.43
7	Null model	1	210.04	21.22
8	VEGCLASS	4	212.66	23.84

HT=habitat type, HSI=habitat suitability index, VEGCLASS=specific age class of aspen, or non-aspen vegetation type. Null model=intercept only.

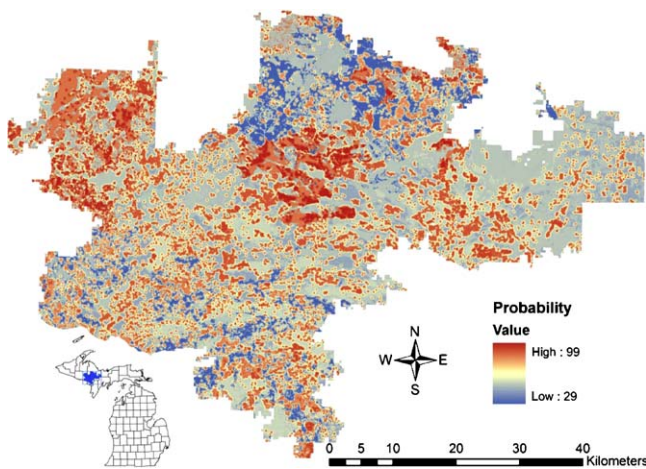


Fig. 3. Spatial distribution of predicted ruffed grouse use of areas for drumming in a 800 km² area in the western Upper Peninsula of Michigan. Data were based on grouse surveys conducted between 2005 and 2006, current land cover, and habitat type classifications.

($SE = 0.47$), $\hat{\beta}_{1,TT} = 0.05$ ($SE = 0.04$), $\hat{\beta}_3 = 0.02$ ($SE = 0.007$). The predictive power of the model was fair with the area under the ROC curve equal to 0.75.

Distinct spatial patterns in grouse drumming habitat use probabilities were evident throughout the study area (Fig. 3). We predicted areas in the central and western portions of the study area, and small areas interspersed throughout, to have high probability of use. These areas corresponded with habitat types on mesic sandy loams or loamy soils (specifically, ATM and AOC), and the hydric TT habitat type.

5. Discussion

Our research presents a method that can be used in conjunction with traditional drumming surveys to relate drumming location to ecological characteristics throughout landscapes. Recording estimated distance and azimuth from the sampling position to the direction where ruffed grouse drumming was heard during spring drumming surveys was a reliable method for determining vegetation type, age class, and habitat type in which grouse were drumming at distances <130 m. At greater distances, the method was 100% reliable for determining habitat type, because patch sizes of habitat types were relatively large (>20% of survey area). The method was 80% reliable for determining vegetation type or age class at estimated drumming distances >130 m. When used in conjunction with GIS, this method is useful for linking spatial and ecological characteristics of landscapes to behavioral aspects of ruffed grouse biology.

Our modeling efforts demonstrated that the top model for predicting ruffed grouse drumming habitat contained the HT and HSI parameters. Other models were not competing, which suggests habitat type and habitat suitability are the best predictors of ruffed grouse drumming habitat. Examination of the parameter estimates shows there was a negative relationship between probability of using areas for drumming and more xeric habitat types (i.e., QAE and AQV). Habitat types characterized by mesic sandy loams or loams, or hydric soils generally had a positive relationship with probability of drumming habitat, with the exception of the AVO habitat type. The relatively large standard errors for model parameter estimates likely reflected the variability observed between years.

The model's predictive power (75%) was reasonable given the many confounding factors in addition to HT and HSI that may contribute to grouse use of an area for drumming. For instance, microsite characteristics (e.g., overstory cover, and woody stem density) may provide suitable structural characteristics at a specific drumming log, whereas data collected at the stand or landscape level may deem an area or HT as generally unsuitable drumming habitat. Additionally, behavioral characteristics such as territoriality or use of primary versus alternate logs (Gullion, 1977) contribute to variability associated with grouse use of areas for drumming. Furthermore, according to the MDNR drumming survey results, there was a significant ($p \leq 0.05$) increase in the number of drums heard per survey route between 2005 and 2006, suggesting a population increase (MDNR, 2006). As the grouse population increases, use trends may differ, and individual grouse may be distributed differently in the landscape. It is likely that the predictive power of the model might be enhanced with longer-term data collected over several years of the grouse population cycle.

Our results supported the literature and previous research that indicates aspen, especially young aspen, is important for providing ruffed grouse drumming habitat in the Great Lakes and northern regions where aspen is prevalent (Gullion, 1977; Kubisiak et al., 1980; McCaffery et al., 1996). For instance, young aspen, in general, had higher suitability than older aspen age classes or other vegetation types (Fig. 2). A compelling result of this study, however, was the obvious differences in the potential of aspen within different habitat types to provide the structure necessary for drumming grouse. Habitat type classifications identify geographic areas with inherent ecological properties (e.g., soil moisture and texture, landform) that indicate the potential for vegetation growth (Coffman et al., 1980; Kotar and Burger, 2000). These inherent properties allow for an understanding of the capabilities of a site to support structural and compositional characteristics suitable for providing ruffed grouse drumming habitat. The HT variable was included in the top 2 models, whereas models that included HSI alone or VEGCLASS (i.e., indicated aspen age classes or non-aspen vegetation types), were ranked lower.

Classifying habitat types is not typically included in habitat suitability analysis for wildlife, but perhaps should be to allow for more effective forest planning. That is, managers may be able to distinguish sites with high potential for providing habitat from those that will likely never provide high or even medium habitat suitability based on inherent abiotic and biotic properties of the site. For instance, across its geographic range, ruffed grouse drumming habitat components include more conifer cover, greater woody stem densities, and greater understory foliage densities (Boag and Sumanik, 1969 [Alberta], Thompson et al., 1987 [Missouri], Fearer and Stauffer, 2004 [Virginia]). These forest structural and compositional characteristics result from soil texture, moisture regimes, landform, or other inherent characteristics of habitat types. Thus, habitat types can be a good predictor of the vegetation structure a site is capable of supporting, and therefore habitat suitability, throughout succession. In cases where

selected habitat types are not likely to provide grouse habitat suitability, forest managers and planners may want to select different forest management objectives for these areas than to try and maintain them for ruffed grouse.

In the top model, habitat suitability was also an important indicator of ruffed grouse drumming habitat. The first 4 variables in the habitat model were structural (Hammill and Moran, 1986), which were inherent properties of habitat types. These variables quantified fall-to-spring cover. The 5th variable, however, quantified interspersed fall-to-spring cover and winter food (i.e., ≥ 0.04 ha of aspen within 305 m of cover [Hammill and Moran, 1986]). This interspersed variable was a function of the current spatial arrangement of vegetation types and not an inherent property of habitat types; thus it was also necessary to predict drumming habitat.

Svoboda and Gullion (1972) stated that in Minnesota, if aspen is not present in an area, breeding ruffed grouse are seldom present. Results of this project suggested that not only is aspen important in the northern regions, but ruffed grouse respond differently to the various ecological conditions that support aspen. Important habitat components, which seem to influence whether or not grouse will use a site for drumming are relatively high shrub or sapling densities (Palmer, 1963; Gullion, 1970; DeStefano and Rusch, 1984), overstory cover (Gullion, 1970), and conifer cover < 1 m (Hammill and Moran, 1986). Stem densities of saplings, conifers, and shrubs 4000–12,000/ha represented suitable conditions for drumming habitat (Hammill and Moran, 1986). In the study area, stem density was $> 12,000$ /ha for old aspen within the AVO and AOC habitat types, and > 4000 within all aspen age classes in the ATM habitat type (Felix, 2008). Conifer cover < 1 m was present within the ATM (consisting of lower branches of white spruce and balsam fir) and TT (consisting of lower branches of swamp conifers) habitat types. Because the ATM, AVO, AOC, and TT habitat types exhibited higher habitat suitability, it is not surprising that they were used more frequently by drumming grouse (Tables 2 and 3, Fig. 2). Svoboda and Gullion (1972) documented greater grouse use of aspen in areas supported by loamy soils and peat than sandy loams and sand; these findings are consistent with our results. The negative parameter estimation for the AVO habitat type, however, was unexpected since the AVO habitat type had the necessary structure and habitat suitability for drumming, and a higher percentage of drumming grouse occurred in the AVO type than did random points (Table 2). This likely occurred because the AVO habitat type was the most available (21.25%) upland habitat type in the study area (Table 1); perhaps there were not enough samples collected to reflect a selection of this habitat type.

The negative association between xeric habitat types (QAE and AQV) and grouse use may be attributed to the dominance of bigtooth aspen or the dry, sandy soils characteristic of these habitat types. Svoboda and Gullion (1972) showed a significant ($p < 0.05$) preference for catkins of quaking aspen over bigtooth aspen during 4 of 8 years of their study, perhaps due to greater energy levels of quaking aspen in mid-April. Additionally, they found grouse use of trees on loamy sand was significantly greater ($p < 0.05$) than use of trees on soils with higher sand content (Svoboda and Gullion, 1972). This trend in grouse use of habitat types also appears consistent in other parts of their range. For instance, in the Appalachians, ruffed grouse preferred vegetation on mesic sites over xeric sites (Fearer and Stauffer, 2004). In Missouri, grouse frequently selected areas with eastern red cedar (*Juniperus virginiana*) present (Thompson et al., 1987), which typically grows best on deep, moist, well-drained alluvial sites (Burns and Honkala, 1990). Grouse in our study area did, however, exhibit some use of vegetation with xeric habitat types (Table 2), and habitat suitability was > 0.5 in some vegetation classes within these habitat types (Fig. 2). Grouse were most likely using those vegetation types because of the red-oak component (Servello

and Kirkpatrick, 1989; McDonald et al., 1998). Red oak occurs after aspen in the successional trajectory of vegetation within the QAE and AQV habitat types (Table 1). Research has, in fact, documented grouse use of oak in Wisconsin (Kubisiak et al., 1980) and in other areas within their range in North America where aspen is absent (Cade and Sousa, 1985; Servello and Kirkpatrick, 1987; Thompson et al., 1987).

6. Conclusions

It is undisputed that aspen is a key to providing good ruffed grouse habitat in the northern parts of their range. What has not been previously investigated, however, is grouse use of aspen (and non-aspen vegetation types) within different habitat types, and the relationship between habitat types and drumming habitat suitability. Additionally, our landscape approach to modeling ruffed grouse habitat and its relationship to forest types is useful in understanding how consequences of forest management will affect the distribution and quality of grouse drumming habitat. We demonstrated that the use of habitat type classifications and maps of the current spatial arrangement of aspen within a planning landscape can predict ruffed grouse drumming habitat and areas that most likely would be used in the spring by drumming grouse. This knowledge can help managers identify and maintain areas with inherent potential to provide drumming habitat through timber harvesting. In habitat types with higher habitat suitability and use in younger aspen age classes, timber harvest may be implemented at 50 years (i.e., the standard rotation age for aspen in the Lake States). In habitat types where habitat suitability and use was higher in older aspen or non-aspen (i.e., AQV, AOC, and TT), perhaps harvest may be deferred to provide unique vegetation features important for grouse and other wildlife species that may use these vegetation classes.

It is challenging for managers to meet multiple forest objectives, but increased knowledge of ecological relationships and wildlife habitat use helps managers better understand potential outcomes of manipulations. This study expanded on existing knowledge of ruffed grouse drumming ecology within the Great Lakes region, and created a useful modeling approach and model for predicting areas likely to be used for drumming. The model provided insight into how grouse use vegetation characteristics and distribute themselves to meet life requisites. The value of modeling wildlife habitat is evident in planning forest management activities and understanding how wildlife may respond to spatial or temporal changes in vegetation types through natural succession or manipulations. Our analysis can help managers understand the breadth of ruffed grouse ecology, establish realistic grouse management goals based on the ecological potential of forest types throughout landscapes to support grouse and aspen, or determine how and where timber harvest may benefit grouse over space and time. Our approach also has utility for analyzing and predicting habitat by other grouse or wildlife species in other geographic regions through the use of developing and applying habitat type classifications for respective planning landscapes. The benefits of using habitat type classifications lies in ability to distinguish variability (and therefore, habitat suitability) within vegetation types that can grow on a wide range of soil texture, moisture, or topographic conditions (i.e., aspen, pine species, oak, hickory, mixed deciduous types), where vegetation type and age class alone would not indicate structure.

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